

Smruti Ranjan Pattanaik

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Career Summary

Third-year Computer Science and Engineering student at Siksha 'O' Anusandhan University (CGPA 9.73), with a strong foundation in full-stack development and applied machine learning research. Currently a Research Intern at IIT Bhubaneswar, building a Byzantine-resilient federated learning framework with dynamic asynchronous aggregation using PyTorch and Flower. Also contributing to open source as a GSSoC 2026 contributor. Built full-stack projects including a rental marketplace and an AI-powered MERN chat application integrating OpenAI's GPT model. Proficient in React.js, Node.js, Express.js, MongoDB, MySQL, and PyTorch, with a strong foundation in Data Structures, Algorithms, and REST APIs. Ranked in the top 2% at university and among 8,600+ participants at the CodeSprint Hackathon. Mentored 15+ junior students in React and DSA, and actively seeking internship opportunities to contribute meaningfully to impactful products.

Experience

Indian Institute of Technology (IIT) Bhubaneswar

Bhubaneswar, Odisha

Research Intern — Office of the SECS

May 2026 – Present

- Working under the School of Electrical Sciences and Computer Science (SECS) on applied machine learning research.
- Designing and evaluating a Byzantine-resilient federated learning framework using PyTorch and Flower, focused on secure and efficient distributed model training.
- Gained hands-on experience in federated systems, adversarial robustness, and asynchronous distributed computing.

GirlScript Summer of Code (GSSoC) 2026

Remote

Open Source Contributor

May 2026 – Present

- Selected as an open source contributor in one of India's largest open source programs with thousands of participants.
- Actively contributing to open-source projects via pull requests, code reviews, and collaborative development on GitHub.

Education

Siksha 'O' Anusandhan University

Bhubaneswar, Odisha

B.Tech in Computer Science and Engineering — CGPA 9.73

2023 – Present

Prananath Autonomous College (State Board)

Khordha, Odisha

Intermediate — 86.66%

2023

Ananta Basudev Bidyapitha (State Board)

Dalak, Odisha

Secondary — 85.66%

2021

Technical Skills

- **Programming Languages:** C, Java, Python, JavaScript, SQL
- **Machine Learning:** Federated Learning, PyTorch, Model Quantization, Distributed Systems
- **Web Technologies:** HTML, CSS, Tailwind CSS
- **Frameworks:** React.js, Node.js, Express.js
- **Databases:** MongoDB, MySQL
- **Tools and Platforms:** Git, GitHub, VS Code, Eclipse, Hoppscotch
- **Core Concepts:** Data Structures and Algorithms, OOP, DBMS, REST APIs

Projects

Byzantine-Resilient Federated Learning with Dynamic Asynchronous Aggregation 2026

Tech Stack: Python, PyTorch, Flower (flwr), Distributed Systems

- Designed a two-stage client screening protocol where clients transmit a quantized (compressed) parameter vector for lightweight server-side validation, sending full parameters only upon verification — reducing transmitted data by 50% while defending against malicious client attacks.
- Built a dynamic periodic asynchronous aggregation strategy with a real-time adaptive time window, adjusting per-round timing based on update volume and arrival skewness — eliminating straggler-related training collapses while preserving Byzantine attack resistance.
- Implemented a confidence-scoring defense using cosine similarity and statistical deviation, achieving 100% detection of simulated sign-flip adversarial attacks with zero false rejections of honest clients.
- Benchmarked against a similarity-based baseline defense on Fashion-MNIST, evaluating bandwidth, computation cost, and detection accuracy across multiple federated learning rounds.

NestAway — Full-Stack Rental Marketplace Platform 2026

Tech Stack: HTML, CSS, JS, Node.js, Express.js, MongoDB, EJS, Bootstrap

- Developed a full-stack property rental platform enabling users to create, view, update, and delete listings.
- Implemented RESTful APIs using Express.js for structured CRUD operations.
- Designed dynamic server-rendered pages using EJS templates. Integrated MongoDB with Mongoose schemas for efficient data storage.
- Implemented authentication and authorization for secure listing management. Built a responsive UI using Bootstrap for seamless cross-device experience.

Converso — AI Chat Application 2026

Tech Stack: MongoDB, Express.js, React.js, Node.js, OpenAI API

- Built a MERN-based conversational AI web application enabling real-time chat with AI-generated responses.
- Implemented REST APIs using Node.js and Express.js to process user prompts securely.
- Designed a responsive React.js chat interface with dynamic message rendering.
- Integrated OpenAI's GPT model and persisted conversation history using MongoDB.

Achievements

- Awarded Merit-Based Scholarship for ranking in the top 2% at Siksha 'O' Anusandhan University.
- Government Scholarships (x2) — Awarded by the state government for academic excellence and performance.
- Completed Infosys Springboard Frontend Web Development certification (2025).
- CodeSprint Hackathon (CodeClash) — Achieved Top 2% ranking among 8,600 participants.
- SIH Ideathon (SOA) — Participated in a national-level hackathon, gaining hands-on experience in problem-solving, teamwork, and rapid prototyping.

Leadership & Community

- Active member of Coder's Guild, guided students in understanding and solving Data Structures and Algorithms (DSA) problems through regular mentoring sessions, improving their problem-solving skills.
- Assisted classmates in mastering DSA concepts and coding problem-solving techniques.